Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahaj-bhadja sahaj-bhadja

Summary

Software engineer with 2+ years of experience building **distributed product systems** across **Java, Python, TypeScript, React, and SQL**, shipping customer-facing APIs, booking-style workflows, and AI-assisted services in microservice architectures with strong attention to reliability and fast production iteration.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

PDEU
B.Tech. in Computer Science and Engineering *Nov 2020 – May 2024*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built and maintained distributed service flows across **FastAPI**, **Azure SQL**, **PostgreSQL**, and cloud storage for a multi-tenant platform serving 500+ users, reducing dashboard latency from 1.2s to 350ms through schema and query optimization.
- Implemented fault-tolerant background jobs and integration workflows for 5K+ monthly tasks with validation, reconciliation checks, and release safeguards, improving throughput 3x and sustaining 99%+ production reliability.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed customer-facing product workflows with **Flask** + **React**, secure file ingestion, and modular APIs, handling 10K+ daily grading requests for 150+ instructors in a production environment.
- Built concurrent processing and release workflows on **AWS RDS** with CI/CD-backed deployments, cutting turnaround time by 60% while maintaining stable API performance under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservices and large-scale API integrations using **Python**, OOP patterns, and **AWS Glue/S3**, improving response times by 30% and pipeline efficiency by 68% in a production codebase.
- Maintained 20+ unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, strengthening release quality and delivery speed in Agile sprint cycles.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based operations platform with **REST APIs**, WebSockets, and microservices, supporting 8+ business workflows and 2M+ synchronized records in a user-facing system.
- Implemented modular auth, billing, and notification services with **Jenkins**, **Docker**, and Azure SQL-backed workflows, reducing deployment time by 50% and sustaining 95%+ successful build validation.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built backend APIs for a self-serve service-booking platform covering booking, reservation tracking, pricing, coupons, and payment-status workflows across 50K+ records with sub-200ms API performance.
- Integrated real-time payment and webhook flows with Firebase synchronization and client-facing booking logic, reducing manual payment handling by 70% while aligning 12+ backend integration points with business requirements.

Skills

Languages: Java, Python, JavaScript/TypeScript, SQL, C++
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, JSON, OOP, Microservices, Background Jobs
Frontend: React.js, Next.js, TailwindCSS, HTML/CSS, WebSockets, State Management
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Relational Data Modeling
Cloud & Infrastructure: AWS (RDS, S3, Glue), Microsoft Azure, Docker, Kubernetes, Jenkins, Git/GitHub
AI/ML: LLM Systems, Prompt Engineering, AI Coding Assistants, Intelligent Automation, Agentic Workflows
Engineering Practices: CI/CD, Unit Testing, Integration Testing, Metrics & Monitoring, Agile, System Design, Performance Optimization

Projects

FinNexus | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built a self-serve fintech product supporting 50K+ users and 1M+ transactions with real-time updates, secure transaction controls, and CI/CD-backed deployment across a scalable distributed stack.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-powered SQL workspace across a **Java** + **TypeScript** codebase, supporting 10+ databases and accelerating query authoring, schema exploration, and developer-facing data workflows.

CPU-Scheduling Simulator | *C++, React.js, JavaScript*
- Implemented 6 scheduling algorithms with interactive **React**-based controls and live visualizations across 1K+ simulated process combinations, improving systems reasoning, performance tuning, and debugging workflows.